

MATH Semester Examination 2024

Mathematics (MATH) · Semester Examination · 2024

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Define Rank of a Matrix. Derive or justify the key properties involved. (5 marks)
2. Write short notes on Mean Value Theorems. (5 marks)
3. Discuss the role of Conditional Probability in Probability & Statistics. (5 marks)
4. Discuss the role of Propositional Logic in Discrete Mathematics & Logic. (5 marks)
5. Define Permutations and Combinations. Derive or justify the key properties involved. (5 marks)

Part B

1. Define Rank of a Matrix. Derive or justify the key properties involved. (Unit 1)
2. Define Systems of Linear Equations. Derive or justify the key properties involved. (Unit 1)
3. Illustrate Mean Value Theorems with a worked example. (Unit 2)
4. Compare Taylor and Maclaurin Series with Partial Derivatives. When would you prefer one over the other? (Unit 2)
5. Illustrate Conditional Probability with a worked example. (Unit 3)
6. What is meant by Bayes' Theorem? Discuss its advantages and limitations. (Unit 3)
7. Explain Propositional Logic with a neat diagram and a suitable example. (Unit 4)
8. Define Predicate Logic. Derive or justify the key properties involved. (Unit 4)
9. Explain Permutations and Combinations with a neat diagram and a suitable example. (Unit 5)
10. Discuss the role of Pigeonhole Principle in Graph Theory & Combinatorics. (Unit 5)